

Static vs. Contextual Embeddings: A Comparative Case Study for NER

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1. MOTIVATION

The purpose of this study is to empirically compare static embeddings (GloVe, Word2Vec) with contextual representations (BERT, RoBERTa). We analyze performance across specific linguistic challenges to identify where modern Transformers excel and where simpler models remain viable.

- **Accuracy Analysis:** F1, Precision, and Recall metrics.
- **Stress Testing:** Polysemy, OOV, and Syntax.
- **Resource Trade-offs:** Efficiency vs. Effectiveness.

2. METHODOLOGY

Pipeline A: Static (BiLSTM)

Word vectors (GloVe 100d, W2V 300d) are fed into a Bidirectional LSTM. This architecture relies on fixed lookups, mapping unseen words to a single <UNK> vector.

Pipeline B: Contextual (Transformers)

Fine-tuned BERT and RoBERTa models. These utilize subword tokenization and self-attention to generate dynamic vectors based on the entire sentence structure.

3. EXPERIMENTAL SETUP

- **Dataset:** CoNLL-2003 (NER benchmark).
- **Hardware:** NVIDIA Tesla T4 GPU (Google Colab)
- **Subsets for Stress Testing:**
 - **Polysemy:** Ambiguous words (≥ 3 tags).
 - **OOV:** Words absent from training vocabulary.
 - **Short vs. Long:** Sequence length impact (< 10 vs. > 25 tokens).

4. QUANTITATIVE PERFORMANCE

F1 Score	GloVe	W2V	BERT	RoB
Overall	0.7377	0.6763	0.8924	0.9078
Polysemous	0.7275	0.6747	0.8885	0.9045
OOV Subset	0.7059	0.6369	0.8899	0.9082
Short (< 10)	0.7602	0.6924	0.9040	0.9031
Long (> 25)	0.7042	0.6466	0.8805	0.9175

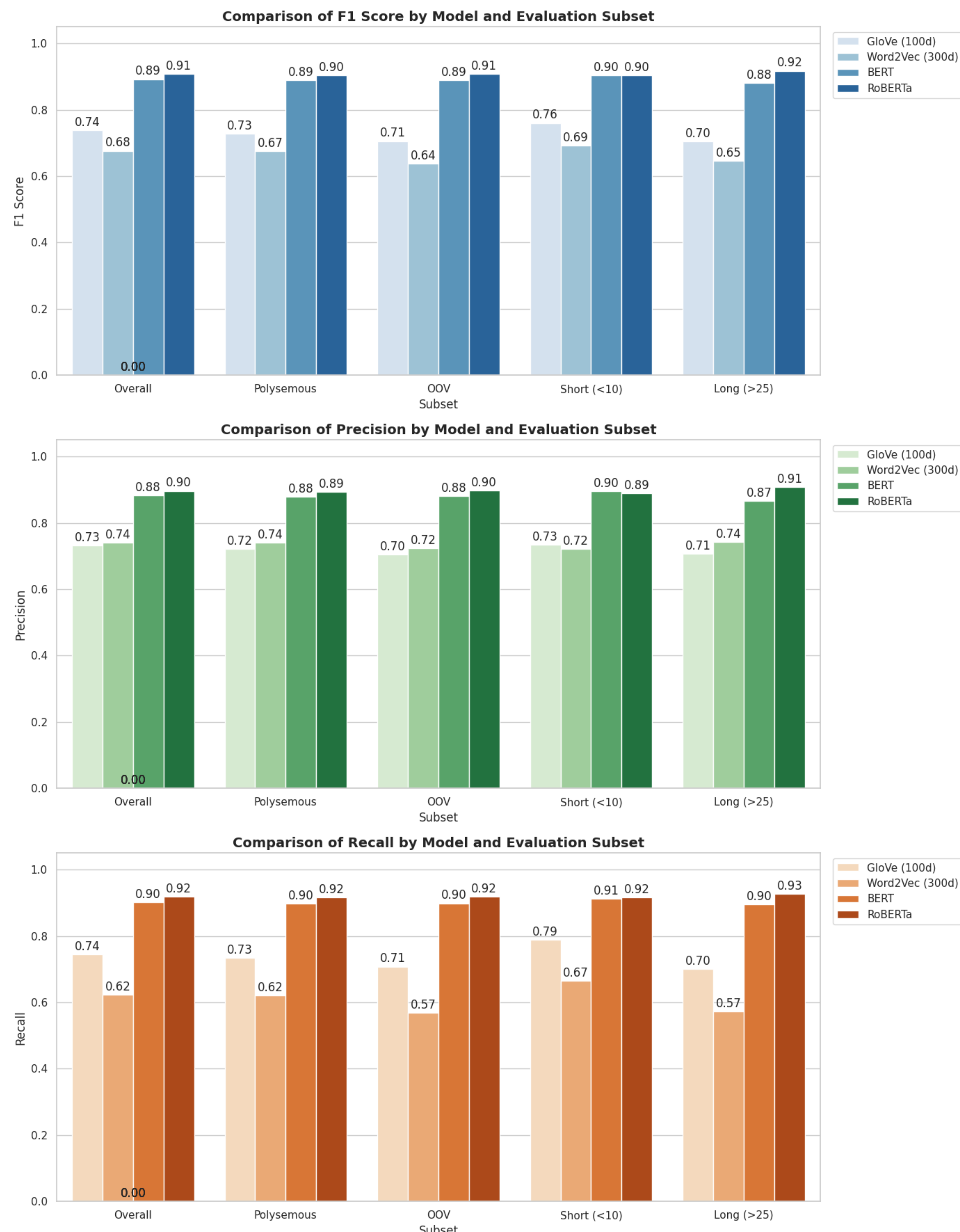


Figure: Performance Comparison: F1, Precision, and Recall trends show RoBERTa and BERT dominating in OOV and Long sequences.

Inference Speed	Avg Latency (ms)
Static (BiLSTM)	0.01 ms
Contextual (Transformer)	0.13 ms

Static models are $\sim 13\times$ faster, highlighting their value in high-throughput edge environments.

5. QUALITATIVE ANALYSIS

Case 1: Polysemy Handling

"Oleg Shatskiku made sure..."

Static models mistag surnames as O, while Contextual models preserve the full span.

Case 2: OOV Recovery

"Uzbekistan are in the finals..."

Static models confuse LOC with PER. Transformers recover via subword features.

Case 3: Long-Range Dependency

Self-attention accurately links distant tokens where LSTMs suffer from information decay.

6. CONCLUSIONS

- Subword Robustness:** Transformers maintain > 0.89 F1 on OOV terms where static models drop to 0.63.
- Precision vs. Recall:** Word2Vec (300d) showed a high precision but a severe recall deficit (0.62 overall), often missing entities entirely.
- Syntactic Stability:** Long sentences (> 25 tokens) cause a $\sim 6\%$ F1 drop in GloVe, while RoBERTa actually improves, showing the power of self-attention over sequential RNNs.
- Short Sequences:** Performance is most comparable on short sequences (< 10), where static models achieve their peak F1 (0.76).

ACKNOWLEDGEMENTS

This study utilized the CoNLL-2003 dataset.